

DIGITAL FORENSIC EXAMINATION REPORT

Forensic Acquisition of a USB Storage Device

Bit-by-Bit E01 Imaging and Hash Verification with AccessData FTK Imager

FTK IMAGER · E01 (ENCASE) · MD5 / SHA-1 VERIFICATION

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EVIDENCE REFERENCE	4 GB USB Flash Drive (PHYSICALDRIVE1), Case No. 001, Evidence No. 001
EXAMINATION FOCUS	Bit-by-bit forensic acquisition and post-imaging integrity verification
PRIMARY TOOLS	AccessData FTK Imager (Windows)
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This document reproduces coursework performed to demonstrate forensically sound acquisition of a physical storage device for digital forensics education, in a controlled lab environment. All steps were independently executed and verified by the author.

Report Details

Field	Detail
Examiner	Awais Ahmed
Role	Digital Forensics Student / Examiner-in-Training
Source Practical	"Creating a Forensic Bit-by-Bit (E01) Image of a USB Drive" worksheet
Evidence Source	4 GB USB flash drive, imaged via AccessData FTK Imager

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1. Executive Summary

This report documents the forensically sound acquisition of a 4 GB USB flash drive using AccessData FTK Imager, as part of Digital Forensics and Cyber Security (DFCS) coursework aligned to Certified Computer Forensics Analyst (CCFA) competencies. It covers the full acquisition workflow end to end: selecting the correct source device, choosing the E01 (EnCase) evidence format, recording case and evidence metadata, configuring the destination and compression settings, running the bit-by-bit acquisition, and verifying the resulting image's integrity through MD5 and SHA-1 hash comparison.

The resulting file, USBImage.E01, was confirmed to be a complete, unaltered copy of the source drive: both computed hashes matched their stored verification values exactly, and no bad blocks were found. This report walks through every step of that process in order, with the reasoning behind each choice made along the way, so the acquisition can be understood and reproduced by anyone reviewing it.

2. Background: Why a Forensic Image, Not a File Copy

An ordinary copy-paste operation only captures the files that are currently visible on a device, and can itself alter file timestamps or metadata in the process, exactly the kind of change an examiner cannot afford to introduce into evidence. A forensic bit-by-bit (bit-stream) image instead captures every sector of the source media, including deleted files, file-system slack space, and unallocated space, while leaving the original device completely unmodified.

Hashing the resulting image, both immediately after acquisition and again on demand afterward, and confirming it matches the hash taken from the source during acquisition, is what proves the image is a perfect, unaltered duplicate. That proof is what makes a forensic image admissible and defensible as digital evidence, rather than merely a convenient backup.

2.1 Evidence Image Formats

FTK Imager supports several forensic image formats. E01 (EnCase Evidence File) was selected for this acquisition because it embeds case metadata and cryptographic hashes directly inside the image file itself and supports compression, which is why it remains the most widely used format in forensic casework.

Format	Notes
Raw (dd)	A pure, unstructured sector-by-sector copy with no embedded metadata or hashes of its own.
SMART	An older ASR Data format, largely superseded in modern casework.
E01 (EnCase)	Embeds case metadata and hash values in the file itself, supports compression; used for this acquisition.
AFF (Advanced Forensic Format)	An open, extensible container format, less universally supported by commercial tools than E01.

3. Examination Environment and Tools

- AccessData FTK Imager (Windows), a free imaging and evidence-preview tool from AccessData/Exterro, used for the full acquisition and verification workflow documented in this report.
- Source media: a 4 GB USB flash drive, presented to Windows as \\.\PHYSICALDRIVE1.

- Destination format: E01 (EnCase Evidence File), compression level 6, balancing output file size against processing time.
- Verification: FTK Imager's built-in MD5 and SHA-1 hashing, computed during acquisition and re-checked in an automatic post-acquisition verify pass.

4. Procedure and Results

4.1 Select the Source Evidence Type

Purpose: Tells FTK Imager what kind of source it is about to acquire from, which determines whether the entire physical device is imaged or only a single logical partition.

In FTK Imager, File → Create Disk Image... opens the source-selection dialog. Since the evidence here is a physical USB drive rather than a single partition on it, Physical Drive was selected, ensuring the whole device, including its full partition table and any unpartitioned space, is captured.

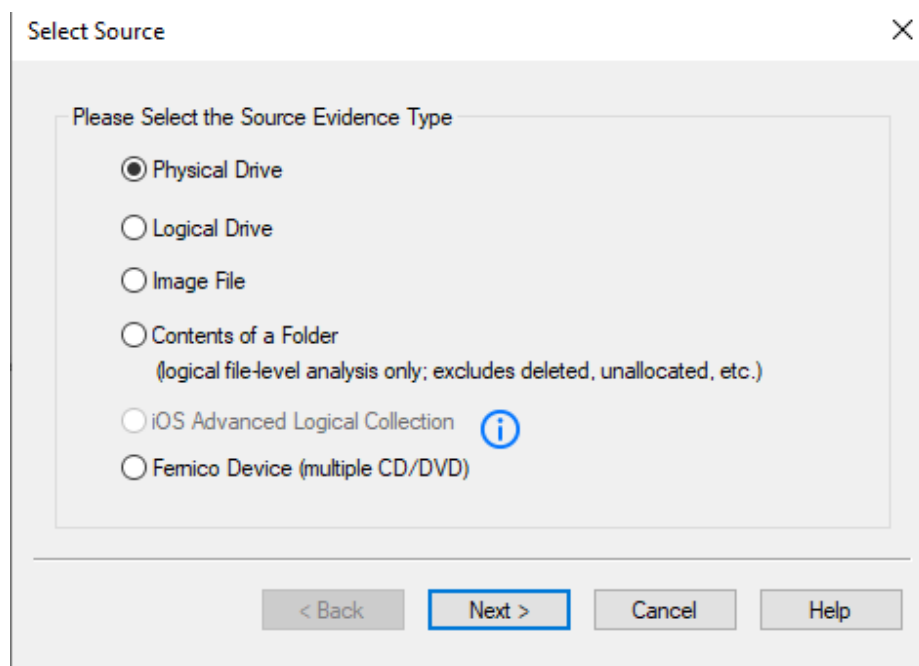


Figure 1. Selecting "Physical Drive" as the source evidence type.

Finding: Source evidence type: Physical Drive.

4.2 Select the Physical Drive

Purpose: Confirms the exact physical device being acquired before any data is read, since acquiring the wrong drive cannot be undone.

FTK Imager lists every physical drive currently connected to the system. The 4 GB USB device appeared as \\.\PHYSICALDRIVE1, clearly identified by its size and interface (SCSI Disk USB Device), which confirmed the correct drive before proceeding.

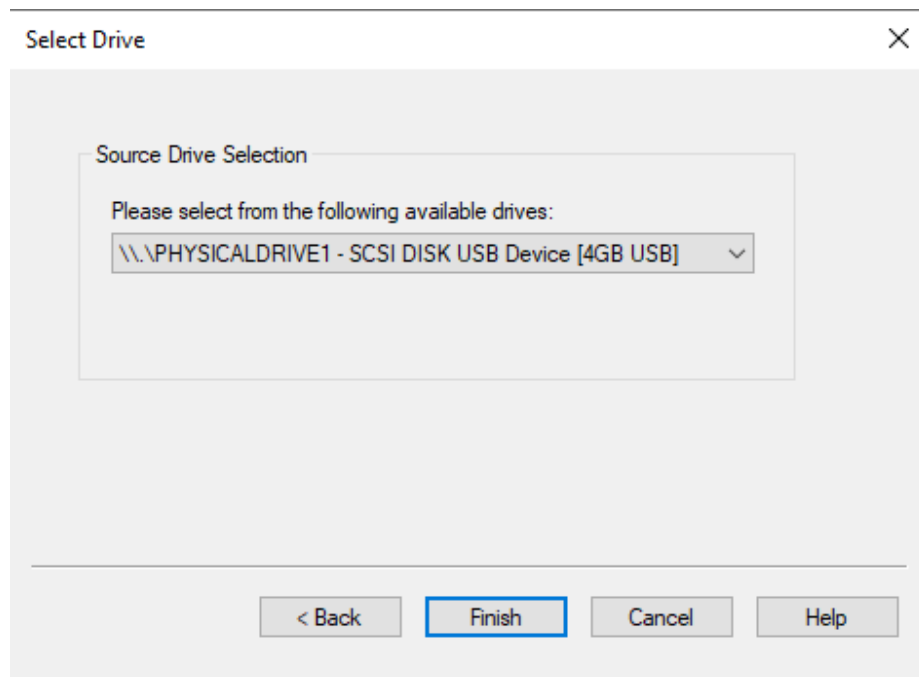


Figure 2. Choosing the correct physical drive from the dropdown list.

Finding: Selected drive: \\.\PHYSICALDRIVE1, SCSI Disk USB Device [4GB USB].

4.3 Choose the Destination Image Type

Purpose: Selects the evidence file format the acquisition will be written in, which determines what metadata and compression options are available.

Of the four formats FTK Imager offers (Raw/dd, SMART, E01, AFF), E01 was selected, for the reasons given in Section 2.1: embedded hashes and case metadata, plus compression support.

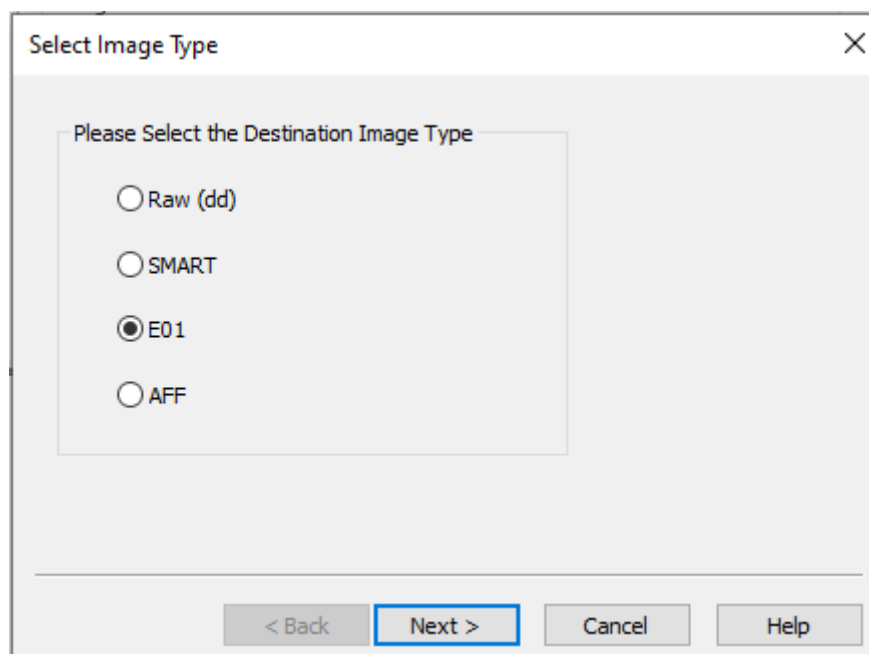


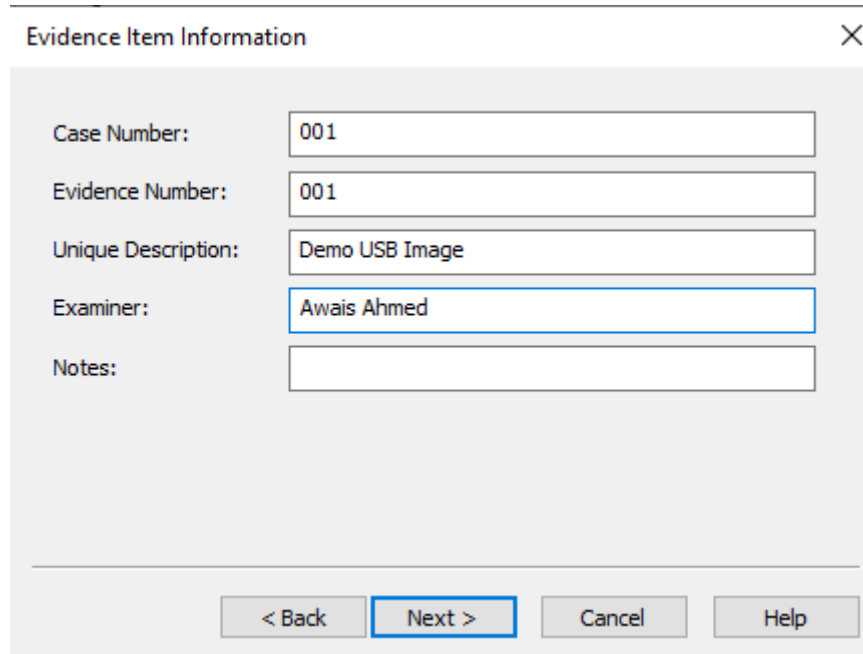
Figure 3. Selecting E01 as the destination image format.

Finding: Destination image type: E01 (EnCase Evidence File).

4.4 Enter Evidence Item Information

Purpose: Records the case metadata that gets embedded directly into the E01 file's header, tying the image to a specific case and examiner from the moment it is created.

Before creating the image, FTK Imager prompts for Case Number, Evidence Number, a Unique Description, the examiner's name, and optional notes. This information is embedded into the E01 file's header, preserving chain-of-custody details alongside the image itself rather than in a separate document that could become disconnected from the evidence.



The screenshot shows a dialog box titled "Evidence Item Information" with a close button (X) in the top right corner. The dialog contains five text input fields, each with a label to its left: "Case Number:" with the value "001", "Evidence Number:" with the value "001", "Unique Description:" with the value "Demo USB Image", "Examiner:" with the value "Awais Ahmed", and "Notes:" with an empty field. At the bottom of the dialog, there are four buttons: "< Back", "Next >", "Cancel", and "Help". The "Next >" button is highlighted with a blue border.

Figure 4. Entering case number, evidence number, description, and examiner name.

Finding: Case Number: 001. Evidence Number: 001. Unique Description: Demo USB Image. Examiner: Awais Ahmed.

4.5 Configure the Image Destination

Purpose: Sets where the finished image file will be written and how it will be split and compressed, balancing manageability and processing time against final file size.

The destination folder and image filename were specified, along with a fragment size of 1500 MB (splitting the image into manageable segments rather than one very large file) and a compression level of 6 (balancing output size against processing time). AD Encryption was left unchecked for this exercise.

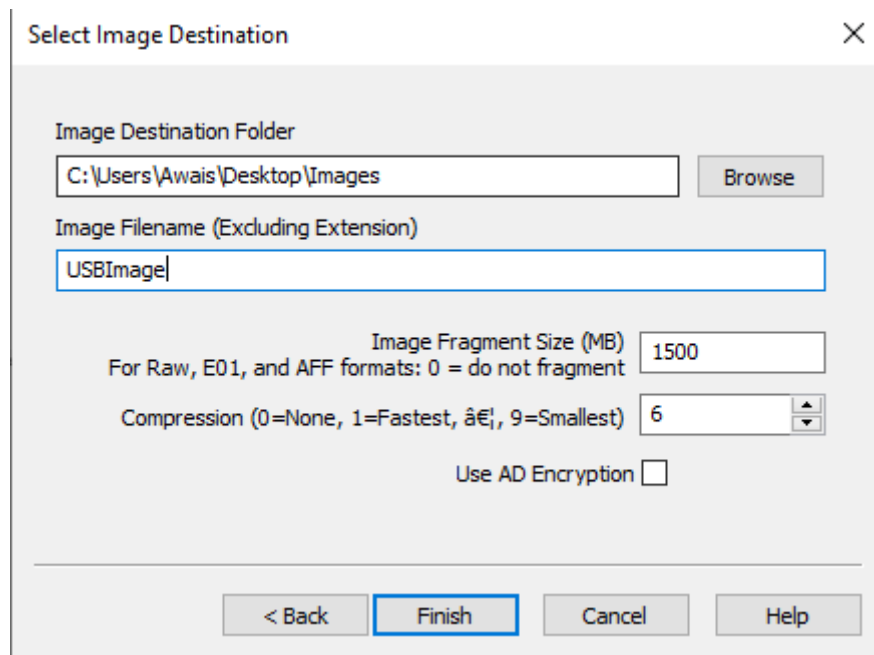


Figure 5. Setting the destination folder, filename, fragment size, and compression level.

Finding: Destination folder: C:\Users\Awais\Desktop\Images. Filename: USBImage. Fragment size: 1500 MB. Compression: 6. AD Encryption: not used.

4.6 Review the Create Image Summary

Purpose: Gives a final chance to confirm the source and destination are correct, and to enable automatic post-acquisition verification, before any data is written.

Before starting, FTK Imager displays a summary of the source and destination, with the option to add additional destinations or overflow locations. “Verify images after they are created” was checked, which tells FTK Imager to automatically re-hash the finished image and compare it against the source once acquisition completes, a critical integrity step covered in full in Section 4.9.

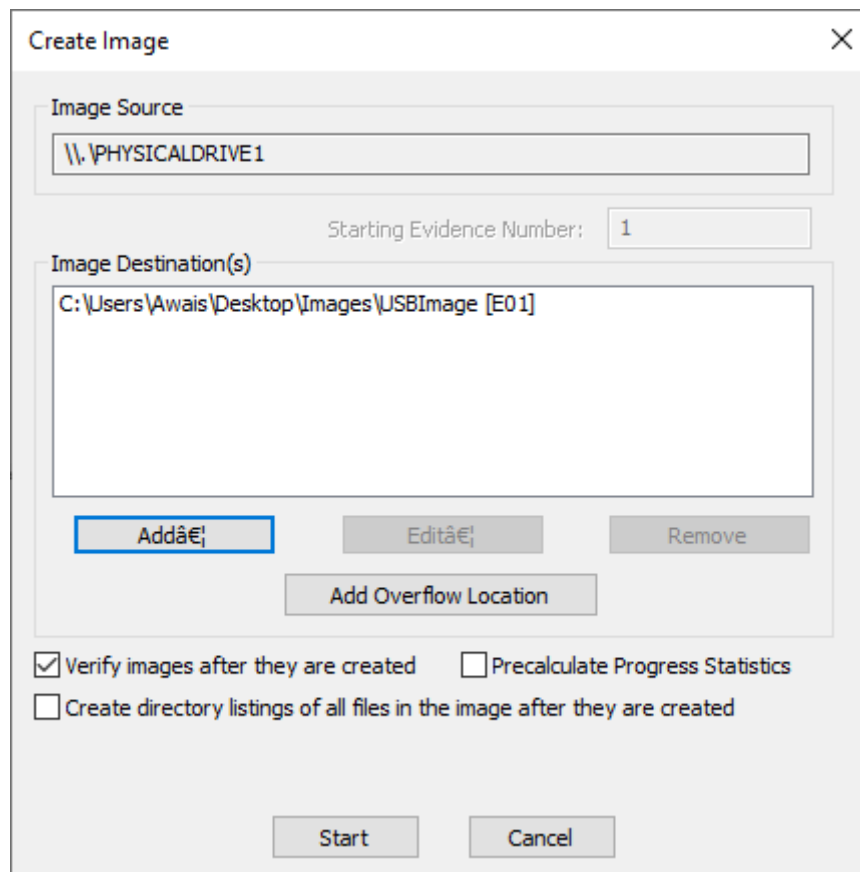


Figure 6. Final review before starting acquisition, with verification enabled.

Finding: Image source: \\.\PHYSICALDRIVE1. Image destination: C:\Users\Awais\Desktop\Images\USBImage [E01]. Verify images after they are created: enabled.

4.7 Acquisition in Progress

Purpose: Documents that the acquisition ran to completion as a normal, monitored process, with live progress rather than an opaque background task.

Once Start was clicked, FTK Imager read the source drive sector by sector and wrote it to the destination E01 file, displaying live progress, elapsed time, and an estimated time remaining throughout.

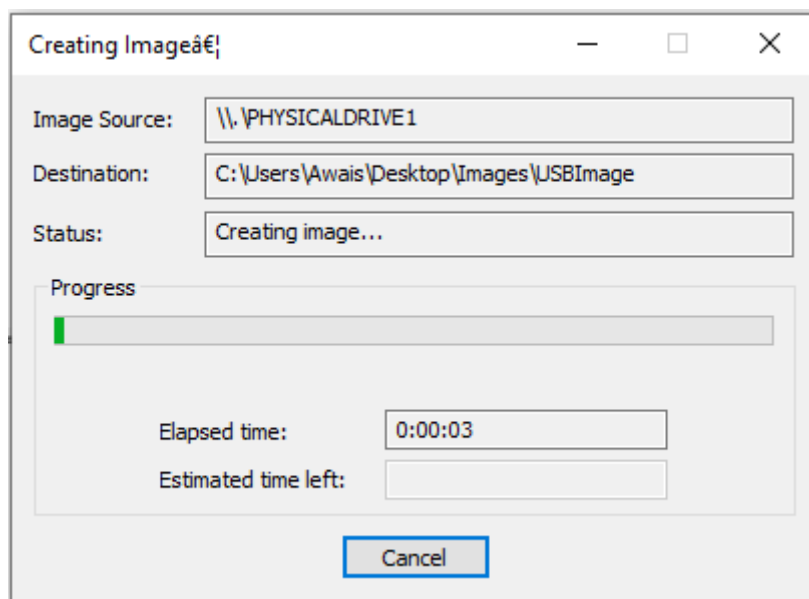


Figure 7. Imaging in progress, with elapsed time tracked in real time.

Finding: Status during acquisition: Creating image...

4.8 Acquisition Complete

Purpose: Confirms the sector-by-sector copy finished without error before moving on to the separate, and equally important, verification step.

When the sector-by-sector copy finished, the status updated to “Image created successfully.” From this dialog, an Image Summary can be opened, or the dialog can simply be closed to move on to verification.

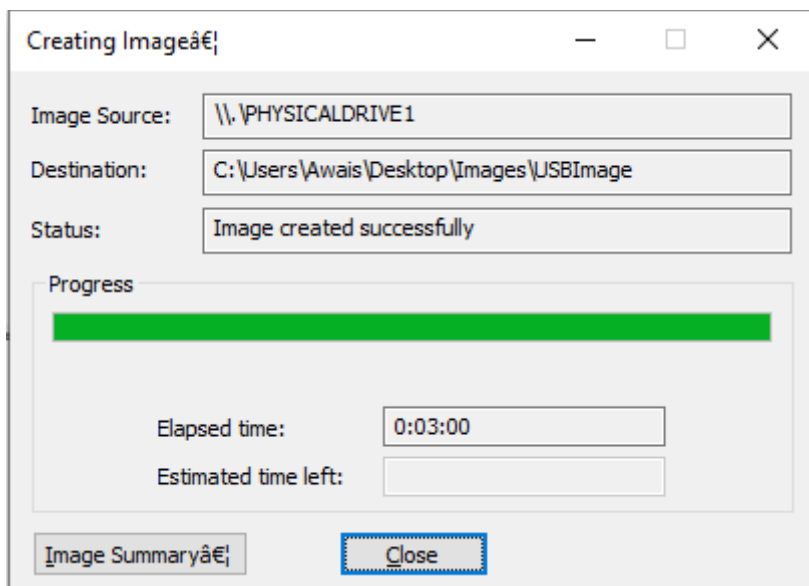


Figure 8. Confirmation that the image was created successfully.

Finding: Status: Image created successfully. Elapsed time: 0:03:00.

4.9 Verify Image Integrity

Purpose: This is the step that actually proves the acquisition is trustworthy: without independent hash verification, a successfully completed copy is not yet forensically proven to be an accurate one.

FTK Imager computed MD5 and SHA-1 hashes of the newly created image and compared them against the hashes computed from the original source drive during acquisition.

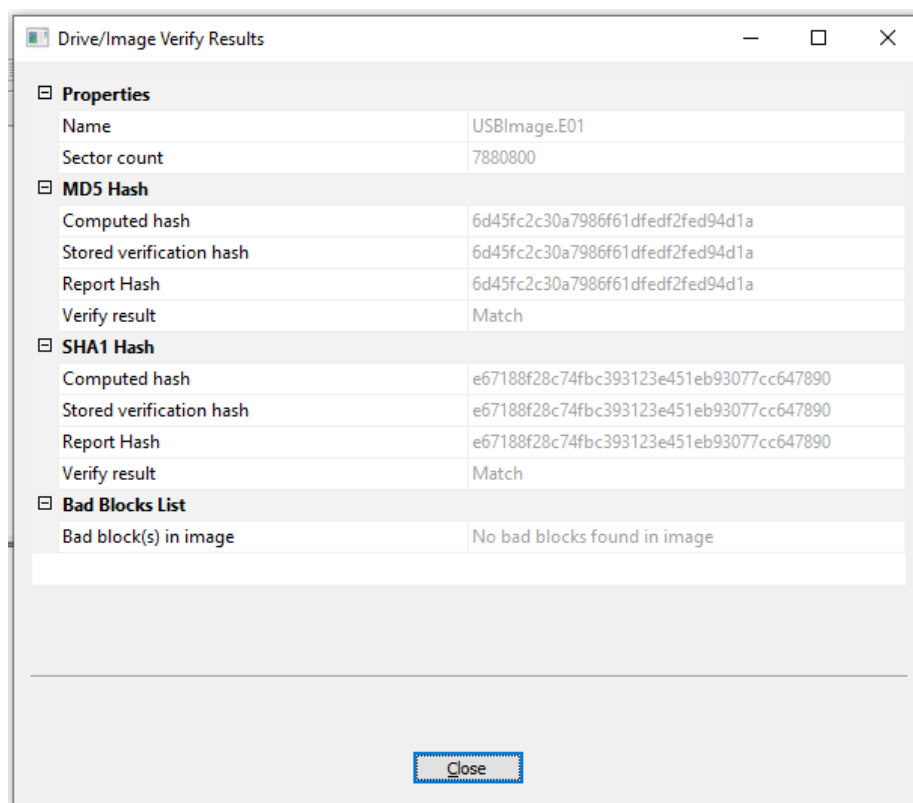


Figure 9. Drive/Image Verify Results showing matching MD5 and SHA-1 hashes.

Finding: Sector count: 7,880,800. MD5 computed hash: 6d45fc2c30a7986f61dfedf2fed94d1a, verify result: Match. SHA-1 computed hash: e67188f28c74fbc393123e451eb93077cc647890, verify result: Match. Bad blocks: none found.

5. Technical Discussion

5.1 What Hash Verification Actually Proves

A matching hash between the source drive and the finished E01 image does not just mean the two files happen to be similar, it means they are bit-for-bit identical. MD5 and SHA-1 are cryptographic hash functions: changing even a single bit anywhere in gigabytes of source data produces a completely different hash value, with no practical way to predict or force a collision. Running two independent algorithms (MD5 and SHA-1) rather than just one further reduces the already-negligible chance that an undetected error could produce a coincidental match in both simultaneously.

5.2 Why This Matters for Admissibility

A forensic image is only as useful as an opposing party's willingness to trust that it accurately represents the original evidence. Recording the acquisition hash at the moment of imaging, then independently re-computing and comparing that hash on demand, gives any later reviewer, not just the original examiner, a way to confirm the image has not been altered since it was created. This is what separates a forensically sound acquisition from an ordinary backup: the backup might be perfectly accurate, but only the hashed, documented acquisition can prove it.

6. Result and Conclusion

TECHNICAL NOTE: RESULT

The resulting file, USBImage.E01, is a verified forensic image of the source USB drive. Both MD5 and SHA-1 computed hashes matched their stored verification hashes exactly, and no bad blocks were found, confirming the image is a complete, unaltered, and court-defensible copy of the original evidence.

6.1 Key Takeaways

- A physical-drive acquisition captures the entire device (allocated data, deleted data, and unallocated space), not just visible files.
- The E01 format embeds case metadata and hash values directly into the evidence file, supporting chain of custody.
- Hash verification (MD5/SHA-1) before and after imaging is what proves forensic integrity; a mismatch would indicate the image cannot be trusted as evidence.
- Documenting each step with case/evidence numbers and an examiner name is standard practice for reproducibility and admissibility.

6.2 Skills Demonstrated

- Forensic disk imaging and evidence acquisition using FTK Imager.
- Understanding of forensic image formats (Raw/dd, SMART, E01, AFF) and when to use each.
- Chain-of-custody documentation (case number, evidence number, examiner).
- Cryptographic hash verification (MD5, SHA-1) for evidence integrity.
- Familiarity with core digital forensics acquisition methodology.